2.5 Proving Statements about Segments and Angles

Name _____ Period ____ Date _____

Geometry 2.5 · Student sheet

I can…

- write a two-column proof.
- name and use the properties of segment and angle congruence.
- complete a proof by supplying statements and reasons.

Vocabulary · fill in the missing word

Proof — a logical argument that uses _____ reasoning to show that a statement is true

Two-column proof — a proof with numbered _____ and their reasons in logical order

Theorem — a _____ that can be proven

Part 1 · Discovery: put the proof in order

Given: A, B, C, D are collinear in that order, and AB = CD. Prove: AC = BD.

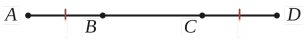
#	Statement	Reason
1		
2		
3		
4		
5		

Part 2 · Notes and examples (fill in each note, then work the example)

Two-column proof: every statement needs a _____: given information, a definition, a postulate, a property, or a theorem.

Example 1

Given: A, B, C, D are collinear in that order, and AB = CD. Prove: AC = BD.



Example 2

Given: M is the midpoint of $\overline{AB}$. Prove: AM = $\frac{1}{2}$AB. Supply the reasons.

Properties of segment congruence: Reflexive ($\overline{AB} \cong \overline{AB}$), Symmetric (if $\overline{AB} \cong \overline{CD}$, then $\overline{CD} \cong \overline{AB}$), and _____ (if $\overline{AB} \cong \overline{CD}$ and $\overline{CD} \cong \overline{EF}$, then $\overline{AB} \cong \overline{EF}$).

Properties of angle congruence work the same way for _____.

Example 3

Name the property.

a. If $\angle 1 \cong \angle 2$ and $\angle 2 \cong \angle 3$, then $\angle 1 \cong \angle 3$.

b. $\overline{PQ} \cong \overline{PQ}$

c. If $\angle A \cong \angle B$, then $\angle B \cong \angle A$.

Example 4

Given: $m\angle ABC = 90°$, D is in the interior of $\angle ABC$, $m\angle ABD = 3x°$, $m\angle DBC = (2x + 10)°$. Prove: $x = 16$.

Part 3 · Partner practice

Partner 1

Name the property: If PQ = QR, then QR = PQ.

Partner 2

Given: L, M, N, P are collinear in that order, and LM = NP. Prove: LN = MP.